
Outcome-Weighted Risk-Aware Supervised Learning for Chinese Standard Mahjong

Improving offense-defense trade-offs under reproducible paired
evaluation

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Baseline architecture: a supervised policy



Structured visible state

Hand, remaining tiles, four rivers and melds, wall progress, shanten, and fan-route cues.

Directly executable policy

One forward pass selects a legal action from 235 slots.

Why naive imitation still deals in

Observed failure

High-cost deal-ins erased gains from otherwise effective offense.

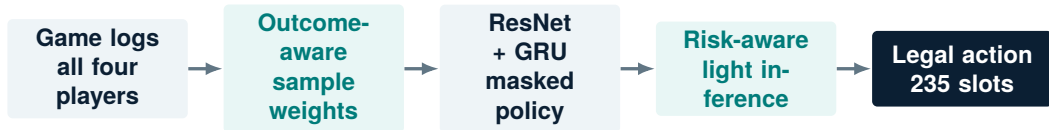
Learning mismatch

Cross-entropy treated every logged action as equally valuable, despite different long-term returns.

The question that drives the redesign

Not only **what** to imitate, but **what deserves more weight**.

Overall framework and three central innovations



1. Outcome weighting

Down-weight low-return and deal-in trajectories.

2. Temporal risk

Track each opponent and the true discard order.

3. Paired evidence

Control walls and seats before comparison.

Outcome weighting reduces low-return influence

Per-player terminal return

$$r_p = \tanh\left(\frac{s_p}{32}\right)$$

Policy weight

$$w_p = w_{\text{fan}} \left(0.20 + \frac{0.80}{1 + \exp(-3r_p)} \right) \times \begin{cases} 0.20, & p \text{ dealt in} \\ 1, & \text{otherwise} \end{cases}$$

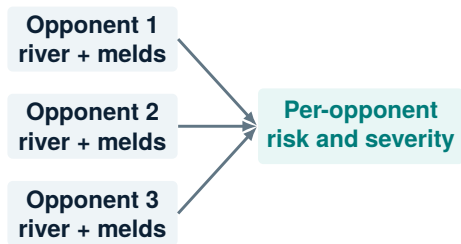
Preserve

Useful defensive choices from non-winners.

Suppress

Low-return trajectories and terminal deal-ins.

Risk is opponent-specific and time-sensitive



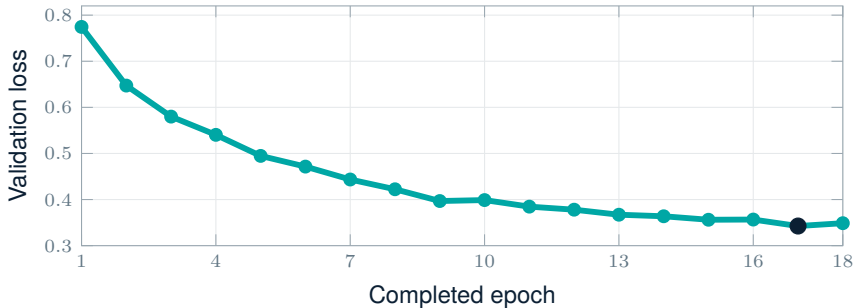
True table-wide order

A GRU retains recency and post-meld evidence.

Light inference

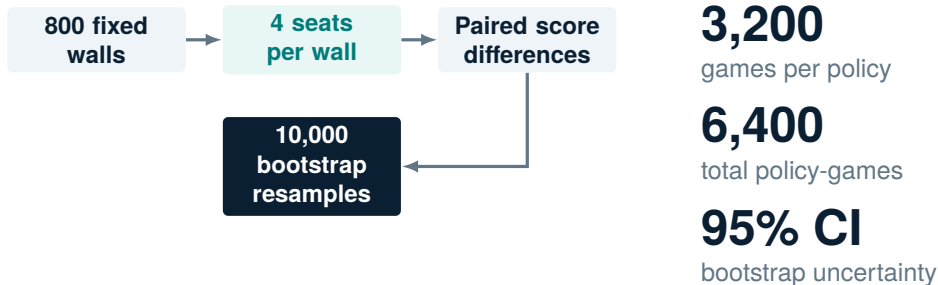
Risk only reranks close discard candidates.

Supervised training loss curve



Validation loss: **0.77** → **0.34** saturation near epoch 17.

Paired evaluation controls wall and seat variance



Primary statistic: same wall, controlled seat, frozen opponents,

$$\Delta_i = \text{score}_{\text{final},i} - \text{score}_{\text{baseline},i}.$$

Outcome weighting improved both score and risk

+1.039

Avg. score (0.49 to 1.53)

+1.19 pp

Win rate (24.53% to 25.72%)

-3.19 pp

Deal-in rate (18.13% to 14.94%)

-0.059

Avg. rank (2.500 to 2.441)



Paired average-score improvement

Win rate rose while deal-in rate fell; the paired CI stayed above zero.

Ablation attributes the gain to the shared policy

Setting	Avg. score	Deal-in
Previous policy	0.934	17.95%
Outcome-weighted policy	1.772	15.70%
+ action-value inference	1.750	15.60%
+ all auxiliary controls	1.796	15.50%

2,000 games per setting; point estimates shown.

Robust contributor

Outcome-weighted training changed the shared policy and produced the consistent gain.

Negative result

Action-value, expected-loss, and fan-route controls lacked reliable incremental evidence.

Competition-tested, with a narrow claim

9th / 16

Competition final

2.37 ms

Mean decision time

Supported conclusion

The evidence supports a **practical, low-cost improvement** of a supervised-learning Bot.

Evaluation boundary. No matched-data state-of-the-art comparison is claimed.

Three takeaways

1. Outcome-weighted imitation

2. Opponent-specific temporal risk

3. Matched-wall paired evidence

Reduce low-return and deal-in noise.

Use opponent identity and discard order to recognize danger.

Control wall and seat variance before policy comparison.

Next

Richer data and augmentation · adaptive aggression · self-play reinforcement learning

References

- [1] K. He, X. Zhang, S. Ren, and J. Sun. “Deep Residual Learning for Image Recognition.” *CVPR*, 2016.
- [2] J. Li et al. “Suphx: Mastering Mahjong with Deep Reinforcement Learning.” arXiv:2003.13590, 2020.
- [3] W. Li et al. “Tjong: A Transformer-based Mahjong AI via Hierarchical Decision-making and Fan Backward.” *CAAI Transactions on Intelligence Technology*, 2024.
- [4] D. Silver et al. “Mastering the Game of Go with Deep Neural Networks and Tree Search.” *Nature*, 529(7587):484–489, 2016.